

Topic Review 1

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March 27, 2025

Schrodinger equation in a central potential $V(r)$.

- Angular momentum, ladder operators
- Combining wave functions and angular momentum.
- Clebsch Gordan tables
- Distinguishable vs indistinguishable in QM (symmetry and anti-symmetry of the wave function)
- spin $\frac{1}{2}$ - fermions

Time independent perturbation theory

- non-degenerate systems (unperturbed system with spectrum $\forall$ energies $\exists$ only one state .
- shift in energy and wavefunction
- first order and second order expansion (careful if $E_k = E_l$ the system wants to select a basis)
- Degenerate energy levels, finding a good basis of the unperturbed system

Variational methods

- Provides an upper bound to the ground state energy, convergence
- parametric trial wave functions, best guess minimises this

WKB approximation

- energy spectra by imposing quantisation conditions relevant to the system

Time dependent perturbation theory

- all the time depends in a phase factor so this isn't apparent in the probability (do simplification before adding perturbations)
- first order ODE coupled or not
- unitary problem $\langle \psi_0 | \hat{V} | \psi_0 \rangle \neq 0$.

Feynman's path integral

- representing the greens function